SYSDESIGN REFERENCE

# Caching Interview Drill

A timed prompt, probing questions, rubric, and answer outline.

Trade controlled staleness and invalidation complexity for lower latency and origin load.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / CACHING

SYSDESIGN

# Caching Interview Drill

**CORE IDEA**

**Trade controlled staleness and invalidation complexity for lower latency and origin load.**

## Reference architecture decision chain

| 01 | 02 | 03 | 04 |
|---|---|---|---|
| Placement | Key Design | Ttl | Eviction |

### Decisions to defend

+ Placement
+ key design
+ TTL
+ eviction
+ consistency
+ and cache-aside vs write-through.

### Failure modes to test

+ Hot keys
+ stampedes
+ stale data
+ low hit rate
+ and oversized values.

## Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

# Caching Interview Drill

| Probe | What a strong answer makes explicit |
| --- | --- |
| How does the design behave during a regional dependency failure? | Requirement, decision, cost, failure behavior, and observable metric. |
| Which metric reveals the first bottleneck? | Requirement, decision, cost, failure behavior, and observable metric. |
| What consistency does the user actually observe? | Requirement, decision, cost, failure behavior, and observable metric. |
| How would the design change at ten times the traffic? | Requirement, decision, cost, failure behavior, and observable metric. |

## Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

## Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

**DESIGN DISCIPLINE**

**For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?**

PUBLISHING STATUS

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